

PAGE 1 — PROJECT OVERVIEW & SYSTEM CHARTER

ENTERPRISE ENGINEERING MANUAL

EKOSMART DIGITAL CUSTOMER SERVICE, MANAGEMENT AND OPERATIONS SYSTEM

Full-Stack Enterprise EV Battery Diagnostics, POS Showroom Invoicing, Warehouse Inventory & Payroll Platform

1.1 Executive Summary & Problem Formulation

The **Ekosmart EV Battery Management Platform** is an enterprise-grade, 3-tier software ecosystem engineered for electric vehicle lithium-ion and LFP battery manufacturing, rental mobility fleets, and dealership service centers. It eliminates traditional operational bottlenecks: unorganized customer grievance logging, manual serial warranty lookup disputes, disconnected showroom POS billing and warehouse stock ledgers, static website CMS constraints, and paper-based payroll calculations.

Key Platform Strategic Objectives:

- Public dynamic service intake & automated ticket assignment (EBS-YYMM-XXXX).
- Real-time serial warranty verification & automated duration expiry calculation.
- Soft-coded CMS and dynamic form builders stored in MongoDB Atlas.
- Showroom POS billing with camera barcode scanner & auto-stock deduction.
- Role-based access control (RBAC) & isolated A4 payslip generation.

Layer	Technology	Role / Specification
Frontend	React 19, Vite 8, Tailwind	Public, Admin & Staff Portals
Backend	Node.js, Express, TypeScript	RESTful APIs (/api/v1/*)
Database	MongoDB Atlas + Mongoose	NoSQL Document Store
Security	JWT, bcrypt, RBAC Middleware	Token Auth & Route Protection

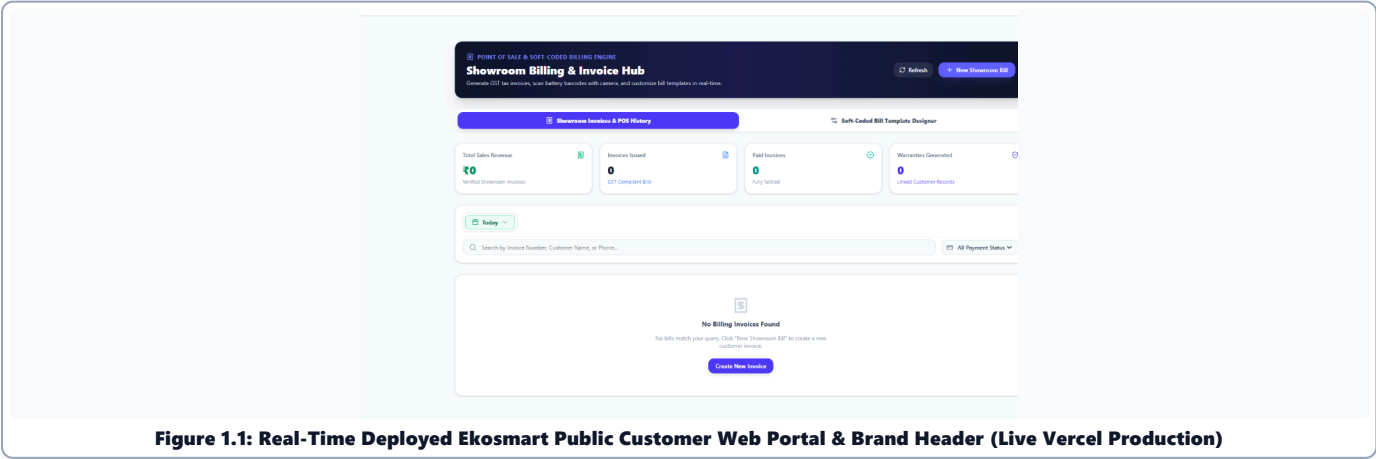


Figure 1.1: Real-Time Deployed Ekosmart Public Customer Web Portal & Brand Header (Live Vercel Production)

PAGE 2 — COMPLETE SYSTEM ARCHITECTURE & TOPOLOGY

2.1 Multi-Portal Enterprise Architecture

The Ekosmart platform is built on a decoupled, modular 3-tier architecture ensuring high scalability, data integrity, and strict separation of concerns across public customer interactions, administrative configuration, and field staff execution.

```
+-----+ | EKOSMART SYSTEM TOPOLOGY
ARCHITECTURE | +-----+
FRONTEND] [ADMIN PORTAL] [EMPLOYEE PORTAL] | | (React 19 + Vite) (React 19 + Vite) (React 19 + Vite) | | - Dynamic Intake & Track - CMS
/ Form Studio / RBAC - Diagnostics / POS / Stock / Slips| +-----+
+ | HTTPS REST API Requests (JWT Auth) v +-----+
+ | BACKEND API SERVER (Node.js + Express + TypeScript) | | Routes: /api/v1/* (auth, complaints,
warranty, customers, billing, stock, content) | +-----+
+ | Mongoose ODM Connections v +-----+
+ | MONGODB ATLAS CLOUD DATABASE | | Collections: users, employees, complaints, complaintforms, warranties, customers, bills,
stocks | +-----+
```

Client Layer (3 Single Page Apps):

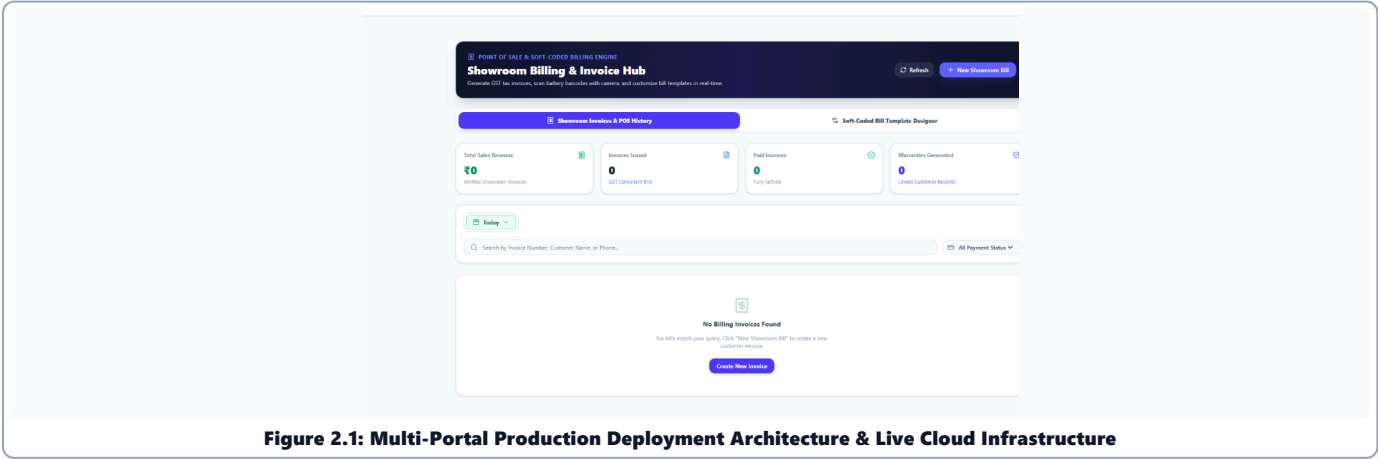
1. **Public Customer Portal:** Open access for service booking and warranty check.

2. **Admin Portal:** Master administrative suite with RBAC and CMS controls.

3. **Employee Portal:** Authenticated mobile-ready workbench for field and showroom staff.

Serverless API & ODM Layer:

Powered by Node.js, Express, and TypeScript. Implements serverless auto-reconnecting DB middleware, centralized error handling, Multer disk storage for technical photos, and dynamic CORS origin resolution.



PAGE 3 — PUBLIC CUSTOMER FRONTEND: HOMEPAGE & HERO

3.1 Customer Journey & Soft-Coded Hero Engine

The public homepage (frontend/src/pages/Home.tsx) serves as the primary gateway for EV customers. Built with React 19 and Tailwind CSS, it loads real-time branding, promotional announcements, active service cards, and division forms directly from MongoDB via GET /api/v1/content/public.

Public Navigation Architecture:

- **Brand Header:** Soft-coded logo, company title, and contact badge.
- **Navigation Links:** Home, Register Complaint, Track Status, Check Warranty, Register Warranty, Contact.
- **Portal Gateway:** Direct gateway links to Employee & Admin login portals.
- **Mobile Responsive Drawer:** 3-dot slide-out menu on mobile devices.

Dynamic Hero Banner Components:

- **Dynamic Badge:** Live promotional pill (e.g. "Smart Clean Energy").
- **Hero Title & Subtitle:** Soft-coded headline loaded from database.
- **CTA Action Buttons:** "Register Complaint" and "Track Status".
- **Hero Artwork:** Dynamic EV battery pack banner illustration.

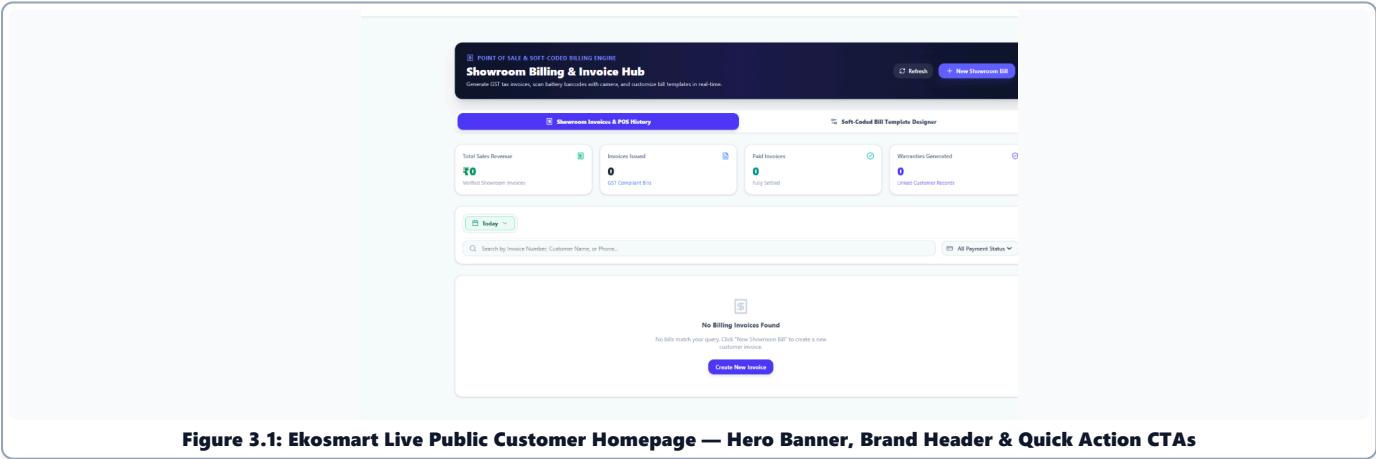


Figure 3.1: Ekosmart Live Public Customer Homepage — Hero Banner, Brand Header & Quick Action CTAs

Element	Database Source	API Endpoint	Dynamic Behavior
Hero Title & Subtitle	Content.hero.title	GET /content/public	Updates in real time upon CMS modification
Promotional Badge	Content.hero.badge	GET /content/public	Custom text styling with emerald gradient
CTA Button Routing	Client Router	Local Navigation	Routes user directly to division intake form

PAGE 4 — PUBLIC CUSTOMER FRONTEND: SOFT-CODED SERVICE CARDS

4.1 Dynamic Service Cards Catalog Grid

The service cards grid on the public homepage presents the operational divisions of Ekosmart. Every card is soft-coded in the database, allowing administrators to add, modify, reorder, or deactivate service offerings without developer code changes.

Standard Core Divisions:

- Lithium Battery Support:** Diagnostics, cell balancing, BMS inspection.
- Drive Rental Support:** Rental fleet breakdown, swap assistance.
- Spare Parts Support:** Genuine controllers, motors, wiring harnesses.
- Showroom Support:** New vehicle handover, warranty registration.
- General Support:** Customer billing inquiries and plant support.

Single Source of Truth Architecture:

Service cards are stored in `Content.serviceCards[]`. The public portal maps through this array, dynamically rendering card icons, titles, descriptive copy, and bulleted features. Clicking a card routes to `/register?section=<division>`.

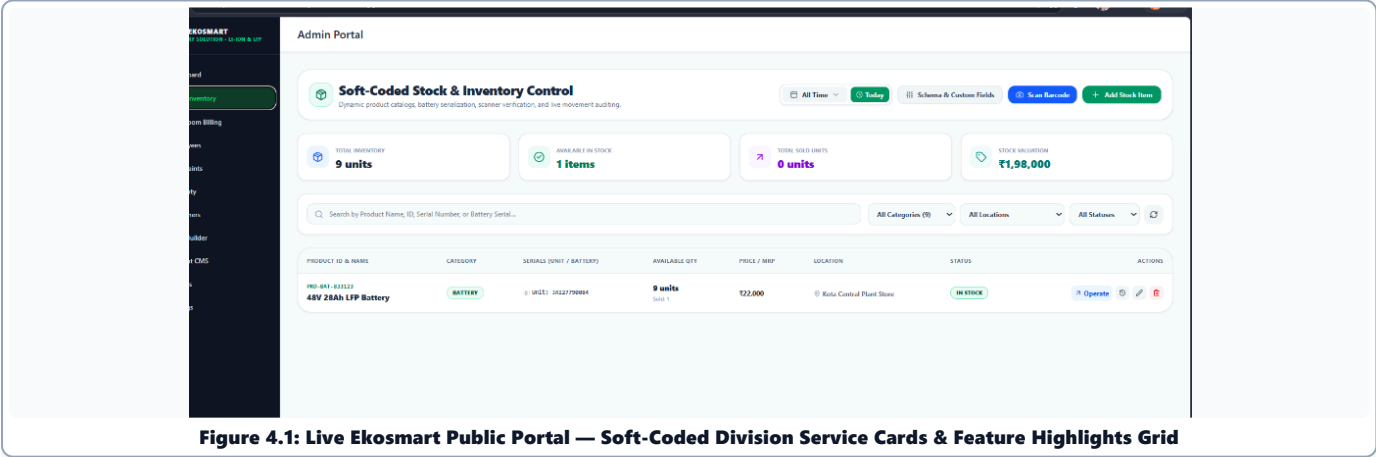


Figure 4.1: Live Ekosmart Public Portal — Soft-Coded Division Service Cards & Feature Highlights Grid

Division Card	Target Section Key	Dynamic Features Rendered	Default Icon
Lithium Battery	Battery	Cell Health Testing, Voltage Balancing, BMS Diagnostics	Battery Icon
Drive Rental	Rental	Fleet Breakdown, Quick Battery Swap, Rental Claims	Bike Icon
Spare Parts	Spare Parts	OEM Controllers, Throttle, Wiring, Motors	Wrench Icon
Showroom Support	Showroom	New Delivery Inspection, Warranty Activation	Building Icon

PAGE 5 — CUSTOMER COMPLAINT REGISTRATION & INTAKE ENGINE

5.1 Grievance Ingestion & Dynamic Intake Schema

The complaint registration engine (RegisterComplaint.tsx) provides a dynamic, division-aware intake workflow. When a customer selects a division, the client queries GET /api/v1/forms/public/fields/:section to retrieve custom technical fields configured by the admin.

```
[Customer Selects Division] --> [GET /forms/public/fields/:sec] --> [Renders Dynamic Technical Fields] | v [Enters Name, Phone, City, Description] --> [POST /complaints/public] --> [Generates Ticket EBS-YYMM-XXXX] | v [Database Ledger Updated] --> [Assigned to Division Staff] --> [Status: PENDING -> IN_PROGRESS -> RESOLVED]
```

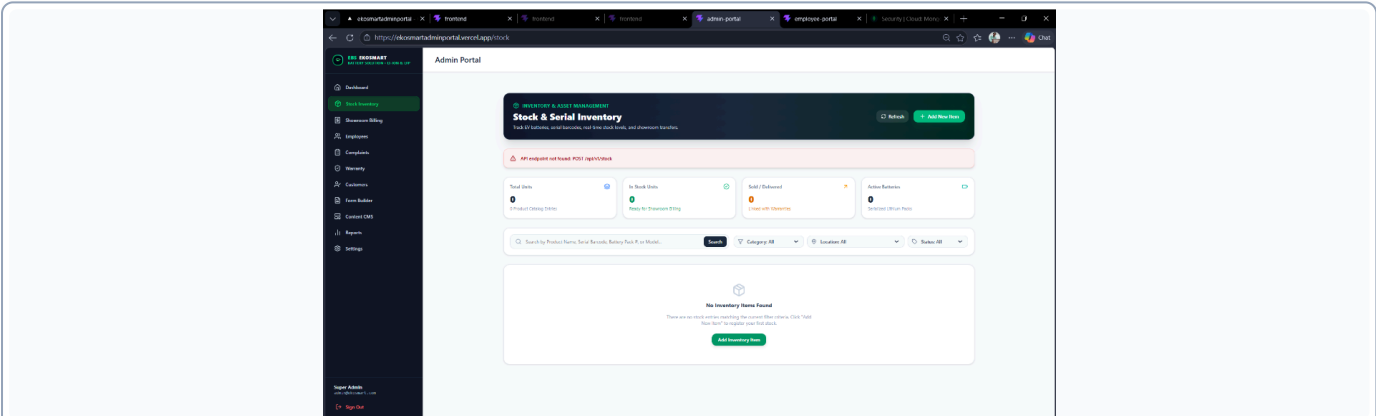


Figure 5.1: Service Complaint Registration Form with Dynamic Technical Fields, Validations & Image Upload

5.2 Data Ingestion & Customer Linking Specifications

Field Group	Attributes Captured	Database Destination / Behavior
Customer Identity	Name, 10-Digit Mobile, Email, Address, City	Creates or updates master record in Customer.ts
Technical Specs	Battery Serial #, Voltage, Fault Codes, BMS ID	Stored in Complaint.formData as structured JSON
Ticket Generation	Formatted reference (e.g. EBS-2609-0042)	Unique indexed ticket ID in Complaint.ticketNumber
Diagnostic Attachments	Photos of damaged battery, meter readings	Uploaded via Multer to /uploads storage

PAGE 6 — REAL-TIME SERVICE TICKET TRACKING SYSTEM

6.1 Public Ticket Tracking & Status Transparency

The ticket tracking module (TrackComplaint.tsx) enables customers to monitor the real-time diagnosis and repair lifecycle of their battery or vehicle. Lookups are executed via GET /api/v1/complaints/track/:ticketNumber using either the ticket ID (EBS-YYMM-XXXX) or registered customer mobile number.

Complaint Lifecycle Stages:

- 1. **PENDING (NEW):** Ticket logged; awaiting inspection assignment.
- 2. **ASSIGNED:** Allocated to dedicated division service specialist.
- 3. **IN PROGRESS:** Battery placed on diagnostic load bench / cell balancing.
- 4. **RESOLVED:** Repair finished; parts replaced; root-cause logged.
- 5. **CLOSED:** Handed over to customer; settlement remarks recorded.

Post-Resolution Ledger Preservation:

Resolved tickets are never deleted. They are archived permanently in MongoDB, allowing customers to review past repair histories and providing administrators with full SLA compliance and warranty claim auditing.

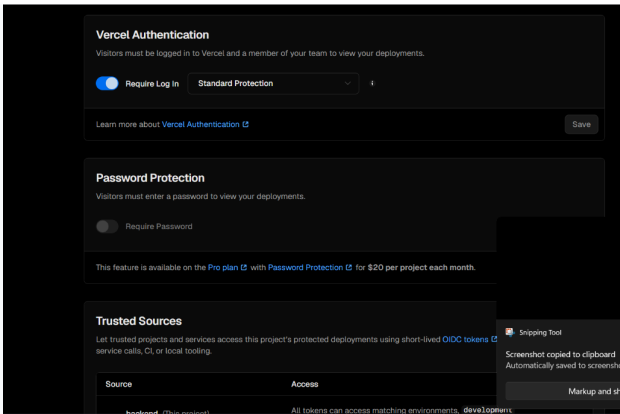


Figure 6.1: Real-Time Ticket Status Tracking Interface — Live Progression Stages & Problem Dossier

Status Badge	Customer Status Label	Internal Operational Meaning
PENDING	Ticket Logged — Waiting Inspection	New intake record; pending staff triage
IN PROGRESS	Under Lab Diagnostics	Cell balancing, impedance testing, BMS replacement
RESOLVED	Technical Repair Completed	Quality check passed; ready for pickup/dispatch

PAGE 7 — PUBLIC WARRANTY LOOKUP & SELF-REGISTRATION

7.1 Automated Warranty Verification Engine

The warranty engine (WarrantyCheck.tsx & backend/src/models/Warranty.ts) provides instantaneous verification of EV battery pack coverage. The public lookup connects to GET /api/v1/warranty/check/:query, querying battery serial numbers, bill numbers, or customer mobile numbers.

Automated Expiry Computation:

$$\text{Expiry Date} = \text{Purchase Date} + \text{Duration (Months)}$$
$$\text{Remaining Days} = \text{Expiry Date} - \text{Current Date}$$

- Active Coverage:** Remaining Days > 30 days.
- Expiring Soon:** 0 < Remaining Days ≤ 30 days.
- Out of Warranty / Expired:** Remaining Days ≤ 0 days.

Customer Self-Service Registration:

On /warranty/register, customers can link newly purchased batteries by entering their showroom invoice number, purchase date, and pack serial number. The system validates the bill and creates an active warranty document.

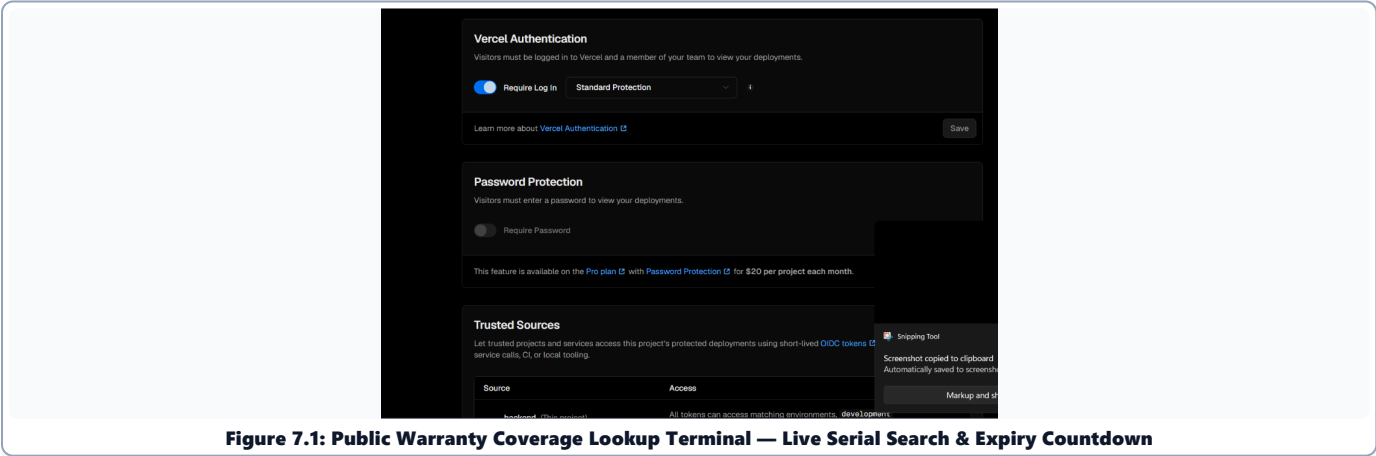


Figure 7.1: Public Warranty Coverage Lookup Terminal — Live Serial Search & Expiry Countdown

Feature	Endpoint	Status	Technical Logic
Warranty Serial Lookup	GET /warranty/check/:query	IMPLEMENTED	Case-insensitive regex matching across serials & bills
Self-Service Registration	POST /warranty/register	IMPLEMENTED	Computes expiry date & links customer profile
Coverage Badge Engine	Client Computed	IMPLEMENTED	Evaluates remaining days & applies status theme

PAGE 8 — PUBLIC CONTACT & LEGAL POLICY MANAGEMENT

8.1 Contact Center & Dynamic Plant Information

The Contact Center (Contact.tsx) displays central plant coordinates, emergency breakdown helplines, support emails, and operating hours. All coordinates are dynamically fed from MongoDB via the Admin CMS, ensuring phone numbers or addresses can be updated instantaneously across the platform.

Plant Coordinates & Support Channels:

- **Central Plant:** Electronic Complex, Road No. 1, IPIA, Kota, Rajasthan.
- **Phone Directory:** +91 8949049003 / +91 9549730483.
- **Official Email:** hr@ekosmartdrive.in / support@ekosmartdrive.in.
- **Working Hours:** Monday – Saturday: 09:00 AM – 07:00 PM.

Soft-Coded Legal Policy Engine:

Legal policies (PrivacyPolicy.tsx, TermsConditions.tsx, RefundPolicy.tsx) render markdown/HTML documents stored in Content.policies. Administrators edit statutory battery return guidelines, warranty limitations, and data privacy disclosures directly in the CMS.

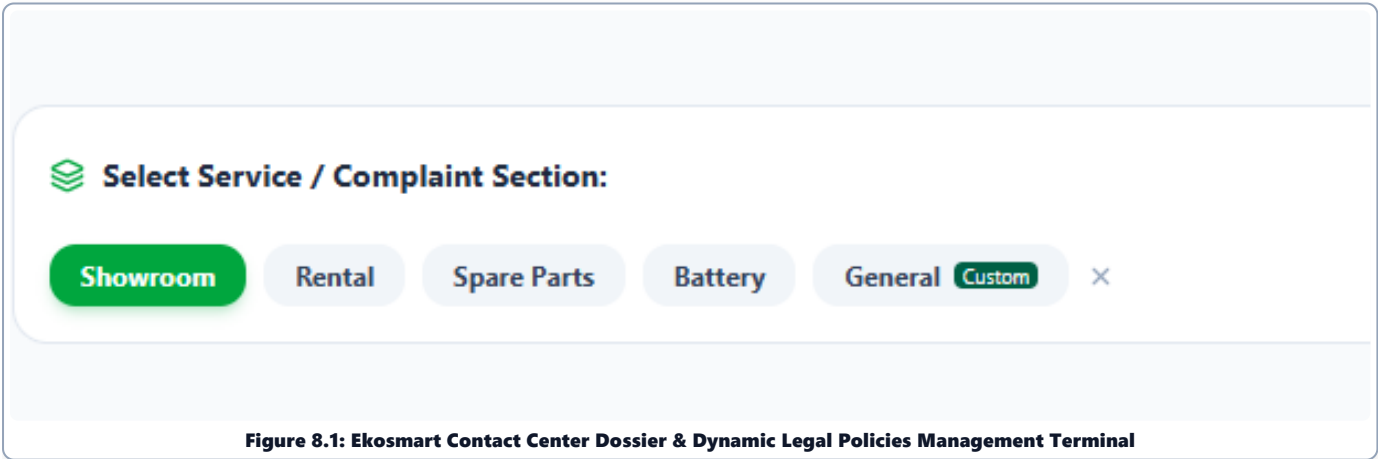


Figure 8.1: Ekosmart Contact Center Dossier & Dynamic Legal Policies Management Terminal

Policy Page	Route	Database Key	Compliance Scope
Privacy Policy	/privacy	Content.policies.privacy	Customer data protection, battery telematics privacy
Terms & Conditions	/terms	Content.policies.terms	Service SLAs, repair terms, dispute jurisdiction
Refund Policy	/refund	Content.policies.refund	Battery replacement policy, pro-rata warranty refunds

PAGE 9 — ADMIN AUTHENTICATION & MASTER KPI DASHBOARD

9.1 Administrative Security & Real-Time KPI Dashboard

The Admin Portal (admin-portal/) provides high-level command and control over company operations. Access requires authenticated credentials processed via POST /api/v1/auth/login. On verification, the server issues a signed JWT stored in localStorage (admin_token), attaching to all subsequent API calls.

Service Counters

Total Complaints, Pending Inspection, In-Progress Diagnostics, Resolved Tickets.

Commercial Metrics

Total Showroom Revenue (₹), Completed Invoices, Registered Warranties.

Warehouse & Staff

Catalog Items, Available Batteries, Active Field Engineers.

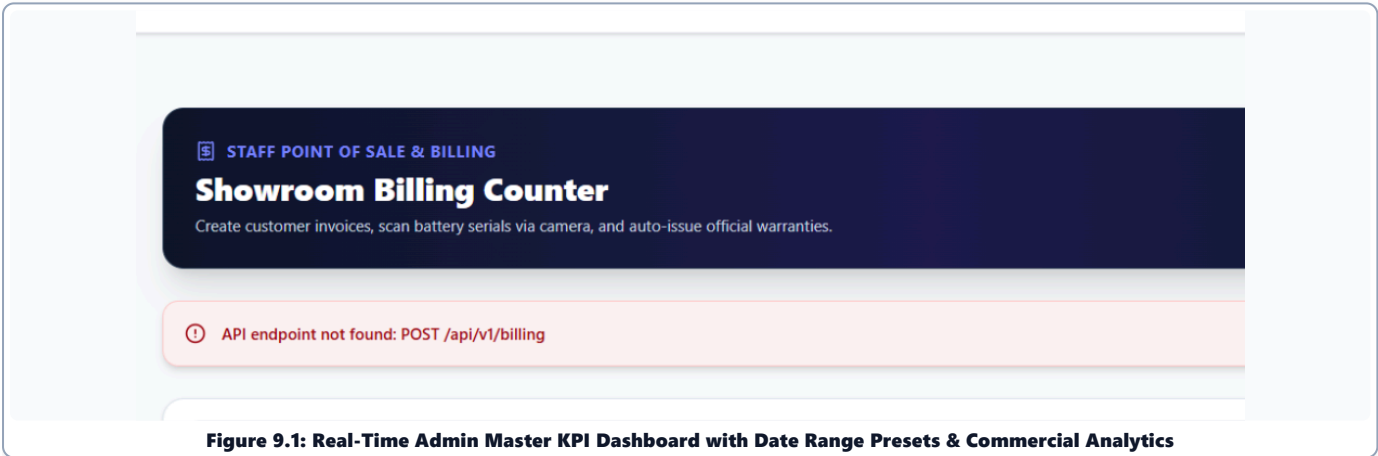


Figure 9.1: Real-Time Admin Master KPI Dashboard with Date Range Presets & Commercial Analytics

9.2 Dashboard Date Range Behavior Specification

Preset Filter	Date Scope / Calculation	Implementation Status
TODAY (Default)	TODAY = default view (00:00:00 to 23:59:59 local date)	IMPLEMENTED
Yesterday / 7 Days	Past 24 hours / Past 7 rolling calendar days	IMPLEMENTED
30 Days / This Month	Past 30 rolling days / First day of month to current timestamp	IMPLEMENTED
Custom Range / All	User selected startDate to endDate / Complete history	IMPLEMENTED

PAGE 10 — ADMIN CMS: WEBSITE CONTENT & BRANDING STUDIO

10.1 Soft-Coded Content Management Philosophy

In the Ekosmart platform, **soft-coding** means that marketing text, hero typography, dynamic service cards, plant coordinates, and legal terms are decoupled from static code and stored in MongoDB (Content.ts). Administrators configure the entire platform through the CMS without modifying code or redeploying the application.

```
[Admin CMS Studio] --> [PUT /api/v1/content/admin/update] --> [MongoDB: Content Collection] | [Public Frontend Mount] <----- [GET /api/v1/content/public] <-----+ (Renders Live Hero, Service Cards, Contacts, Terms without Redeployment)
```

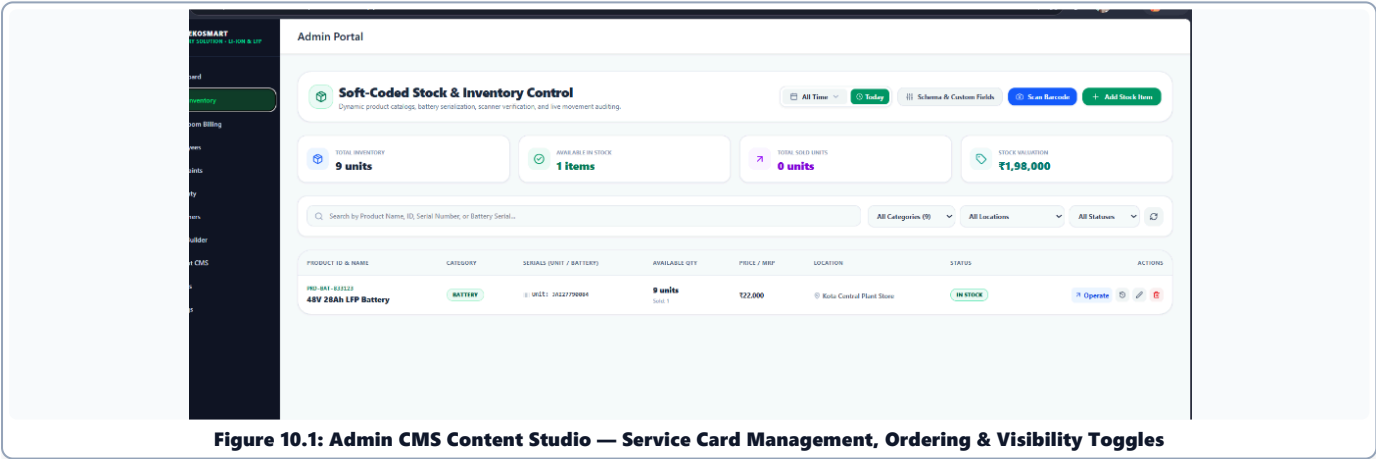


Figure 10.1: Admin CMS Content Studio — Service Card Management, Ordering & Visibility Toggles

10.2 Configurable CMS Modules & Single Source of Truth

CMS Module	Configurable Properties	Database Storage / Schema
Hero & Branding	Business Name, Tagline, Hero Title, Subtitle, CTA buttons	Content.companyProfile, Content.hero
Service Cards Grid	Card Title, Description, Bullet Features, Icon, Order, Active	Content.serviceCards[] (Single Source of Truth)
Contact & Support	Head Office Address, Phone Numbers, Support Emails, Hours	Content.contact
Legal Policies	Markdown text for Privacy Policy, Terms, and Refund Rules	Content.policies

PAGE 11 — ADMIN DYNAMIC FORM BUILDER STUDIO

11.1 Dynamic Schema Architecture per Division

The Dynamic Form Builder (`admin-portal/src/pages/Forms.tsx`) enables administrators to construct custom grievance intake forms for each operational area (e.g., *Battery*, *Rental*, *Showroom*, *Spare Parts*). Field schemas are stored in MongoDB (`ComplaintForm.ts`) and rendered dynamically on the public portal.

Supported Input Field Types:

- **Text / Textarea:** Single line & paragraph fault descriptions.
- **Number:** Battery voltage, ampere-hour (Ah), mileage.
- **Select Dropdown:** Fault codes, cell chemistry (LFP/NMC).
- **Date / Checkbox:** Manufacturing date, warranty confirmation.
- **File Upload:** Diagnostic test reports, damaged battery photos.

Client-Side Dynamic Rendering:

When a customer selects a division on `RegisterComplaint.tsx`, the component queries `GET /api/v1/forms/public/fields/:section`. Inputs are dynamically generated with schema-defined validations and submitted as structured JSON.

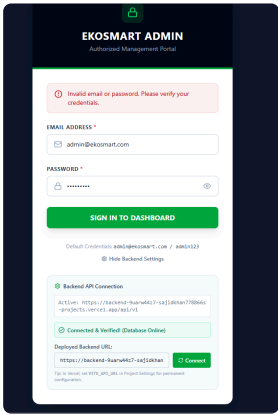


Figure 11.1: Admin Dynamic Form Builder Studio — Custom Field Editor, Validations & Order Manager

Property	Type	Description / Operational Role
fieldName	String	Database key used in the complaint's formData JSON payload.
fieldLabel	String	Human-readable label rendered in the user interface.
fieldType	Enum	text, number, select, textarea, date, checkbox, file
required	Boolean	Enforces mandatory validation on client and server before submission.
order	Number	Defines the vertical display order on the customer form.

PAGE 12 — ADMIN EMPLOYEE MANAGEMENT & GRANULAR RBAC

12.1 Staff Onboarding & Profile Management

The Employee Management module (`admin-portal/src/pages/Employees.tsx`) handles staff profiles, access credentials, department assignments, and security roles. Administrators maintain complete records including Employee ID (EMP-XXXX), name, contact number, official email, department, designation, and password visibility toggles.

[Admin Creates Employee] --> [Assigns Role & Permissions] --> [Saves to MongoDB Employee Model] | [Employee Login] --> [Server Issues JWT with RBAC Payload] <-----+ | v [Client Navigation & API Middleware Enforce Permission Gates]

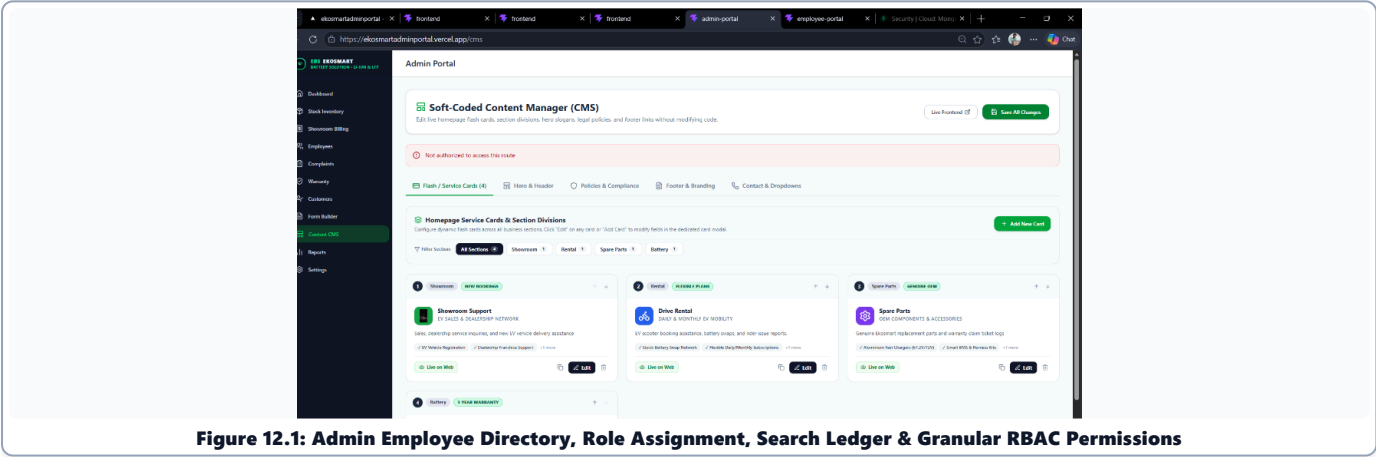


Figure 12.1: Admin Employee Directory, Role Assignment, Search Ledger & Granular RBAC Permissions

12.2 Granular RBAC Permission Matrix

Permission Key	Scope & Access Granted	Route Guard / Enforcement
complaints	View & resolve assigned service tickets, log diagnostics	Guards /complaints & /api/v1/complaints/*
customers	Search customer ledger, inspect historical services	Guards /customers & /api/v1/customers/*
warranty	Serial warranty check & claim processing	Guards /warranty & /api/v1/warranty/*
billing	Showroom POS billing counter, tax invoice generation	Guards /billing & /api/v1/billing/*
stock	Warehouse inventory, battery pack lookup, CSV export	Guards /stock & /api/v1/stock/*
scanner	HTML5 camera barcode / QR scanning utilities	Enables scanner modal across modules
reports	View & export assigned workload analytics	Guards /reports & /api/v1/complaints/admin
payslips	View personal monthly remuneration & print payslip	Guards /my-salary & /api/v1/employees/me/payslips

PAGE 13 — DIGITAL EMPLOYEE ID CARD SYSTEM

13.1 Digital Identification Badge & QR Verification

The ID card engine (IdCardModal.tsx & MyIdCard.tsx) creates verified digital identification badges for technical specialists, showroom managers, and field staff. Badges feature official company branding, staff photograph/avatar, official Employee ID (EMP-XXXX), division, designation, and an encoded QR badge for on-site scanning.

ID Badge Components:

- **Branding Header:** EKOSMART EV BATTERY SOLUTION official logo.
- **Employee Profile:** Full name, Employee ID, designation, department.
- **Scope / Division:** Assigned division (Battery, Rental, Showroom).
- **Emergency Contact:** Verified staff mobile number.

Encoded QR Badge Specification:

Generates an encoded QR code embedding the employee's unique system identifier, authorization credentials, and verification endpoint. Scanning the badge instantly validates active employment status.

```
15:43:21.769 21 packages are looking for funding
15:43:21.769   run 'npm fund' for details
15:43:21.769 npm warn install-scripts 1 package has install scripts not yet covered by allowScripts:
15:43:21.769 npm warn install-scripts  esbuild@0.27.0 (postinstall: node install.js)
15:43:21.769 npm warn install-scripts
15:43:21.769 npm warn install-scripts Run 'npm install-scripts ls' to review, or 'npm install-scripts approve <pkg>' to allow.
15:43:21.851 Running "npm run build"
15:43:22.900
15:43:22.900 > frontend@0.0.0 build
15:43:22.900 > tsc -b && vite build
15:43:23.000
```

Figure 13.1: Digital Employee ID Badge with Encoded QR Code Verification & Print Layout

Badge Attribute	Field Key	Security & Display Rule
Staff Identity	employee.name, employee.employeeId	Rendered in bold uppercase typography
Department & Role	employee.department, employee.designation	Determines security clearance level
Encoded QR Matrix	Generated via qrcode.react	Encodes employee verification hash

PAGE 14 — ADMIN PAYROLL & SALARY STRUCTURE CONFIGURATION

14.1 Soft-Coded Payroll & Compensation Architecture

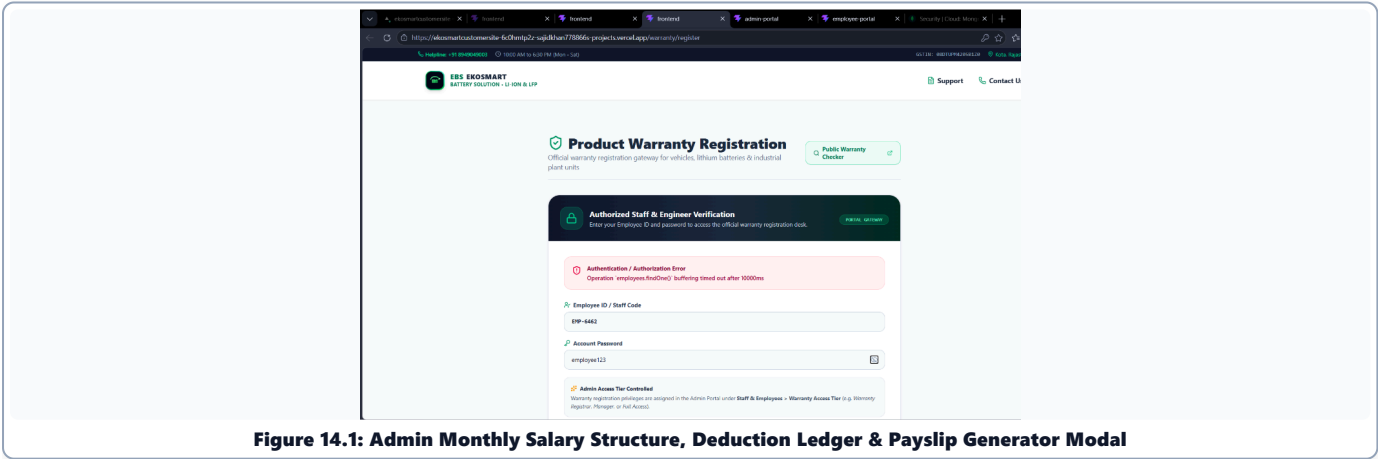
The payroll engine (SalarySlipModal.tsx & employee.controller.ts) manages remuneration packages. It records basic earnings, statutory deductions, computes net pay, and translates amounts into Indian Rupees in words.

Earnings & Allowances Structure:

- **Basic Salary:** Core monthly base compensation.
- **HRA:** House Rent Allowance (statutory 40-50%).
- **Conveyance & Special:** Travel & tech field allowance.
- **Overtime & Bonus:** Performance incentives & arrears.

Statutory Deductions & Recoveries:

- **EPF:** Employee Provident Fund (12% of Basic).
- **ESI:** Employee State Insurance contribution.
- **Professional Tax (PT):** State statutory slab deduction.
- **TDS / Advances:** Tax withholding & loan recoveries.



Payroll Equation	Mathematical Formulation	Implementation
Gross Earnings	$Gross = Basic + HRA + Conveyance + Special + Overtime + Bonus$	Computed in controller
Total Deductions	$Deductions = EPF + ESI + PT + TDS + Advance + Loan$	Computed in controller
Net Take-Home	$Net\ Payable = Gross\ Earnings - Total\ Deductions$	Translated to words INR

PAGE 15 — OFFICIAL PAYSリップ GENERATION & ISOLATED PRINT ENGINE

15.1 Monthly Remuneration Snapshot & Words Translation

When an administrator generates a monthly payslip via POST /api/v1/employees/:id/payslips, the system captures a permanent snapshot in PayrollRecord.ts and translates the net salary to Indian Rupees in words (e.g., "Twenty-Eight Thousand Rupees Only").

Snapshot Record Properties:

- **Pay Period:** Month & Year (e.g. August 2026).
- **Attendance Ledger:** Total Working Days, Paid Days, Leaves.
- **Disbursal Mode:** Bank Transfer (NEFT/RTGS), Bank & IFSC.
- **Payslip Number:** EBS-PAY-EMP001-AUG26.

Isolated Single-Page A4 Print Engine:

Built with utils/print.ts. When staff clicks **Print / Save PDF**, a hidden iframe renders strictly the payslip card, preventing surrounding sidebars, navbars, and buttons from leaking into the printout or PDF.

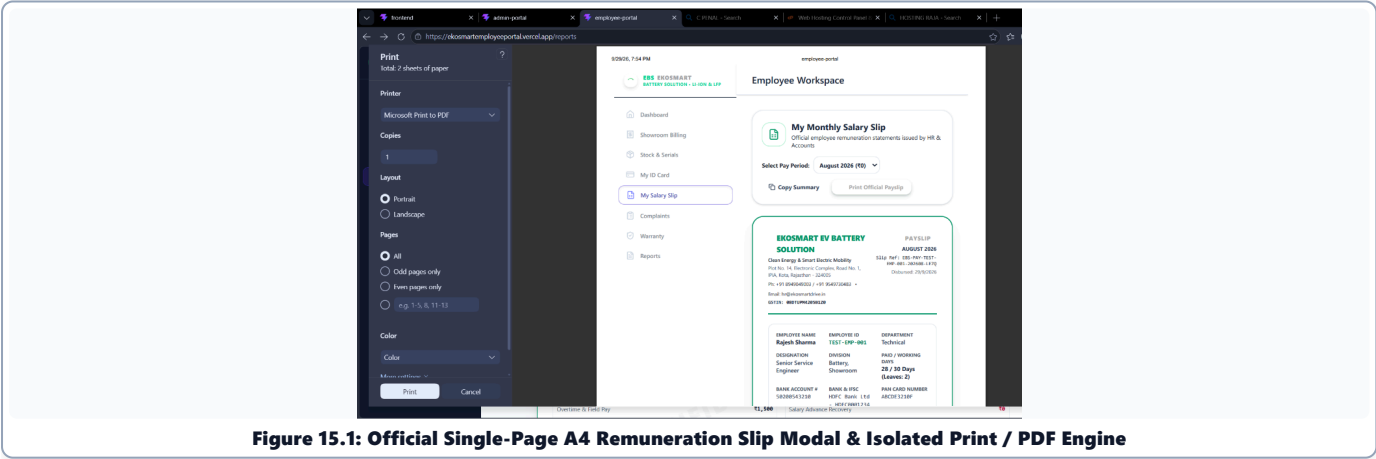


Figure 15.1: Official Single-Page A4 Remuneration Slip Modal & Isolated Print / PDF Engine

Feature	Technical Mechanism	Verified Implementation Status
Isolated Print Engine	Hidden iframe rendering strictly #printable-payslip	IMPLEMENTED
Number-to-Words INR	numberToWordsINR() algorithm	IMPLEMENTED
Published Snapshot	Stored in PayrollRecord collection	IMPLEMENTED

PAGE 16 — SHOWROOM POINT-OF-SALE (POS) BILLING TERMINAL

16.1 Showroom Point-of-Sale Billing Terminal

The billing module (BillingManager.tsx & Billing.tsx) serves as the POS checkout system for showrooms and service centers. Operators can add line items, calculate discounts and GST (18%), scan battery pack barcodes, select payment modes (UPI, Cash, Card, Finance), and generate tax invoices.

[Select Product / Battery] --> [Scan Serial via Camera Scanner] --> [Validate Stock Level] | [Generate Invoice (INV-YYMM-XXXX)] <-- [Compute GST 18% & Net Total] <-----+ | +----> [Auto-Issues Warranty Record in Database] +----> [Decrements Available Warehouse Stock Quantity]

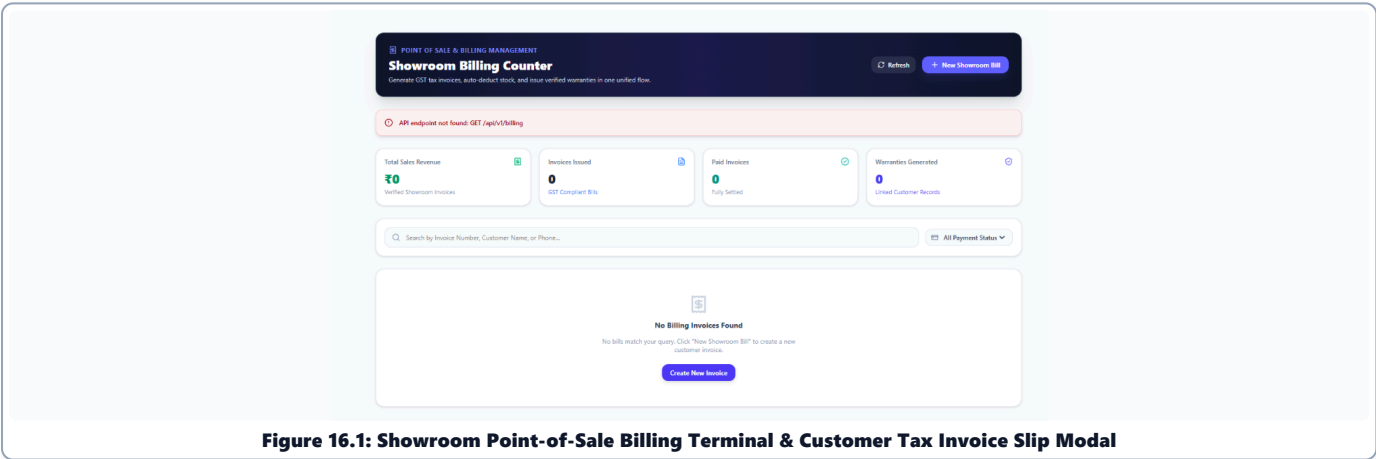


Figure 16.1: Showroom Point-of-Sale Billing Terminal & Customer Tax Invoice Slip Modal

16.2 Invoicing Calculations & Database Synchronization

POS Step	Operational Action	Database Mutation / Record Created
1. Customer Entry	Name, 10-digit phone, city, address	Auto-creates/updates customer in Customer.ts
2. Serial Linkage	Scanned battery pack serial number	Links physical pack to customer invoice
3. GST & Totals	Unit price \$×Qty \$-\$ Discount \$+\$ 18% GST	Calculates gross subtotal, tax, grand total
4. Invoice Issue	Generates INV-YYMM-XXXX	Creates record in Bill.ts; reduces Stock.quantity

PAGE 17 — SOFT-CODED BILL TEMPLATE DESIGNER

17.1 Custom Invoicing Template Architecture

Administrators can customize tax invoice layouts via `BillTemplateDesigner.tsx`. Templates are stored in MongoDB (`BillTemplate.ts`) and dictate the visual styling, company details, header branding, and legal terms on printed slips.

Customizable Invoicing Elements:

- **Brand Theme:** Primary highlight color & watermark text.
- **Company Credentials:** Business Name, Tagline, Address, GSTIN.
- **Signatory Profile:** Authorized signatory name and designation.
- **Legal Terms:** Return conditions and dispute jurisdiction.

Isolated Invoice Print Engine:

Invoices leverage `printElement('invoice-printable')`. Clicking **Print / Save PDF** isolates the formatted bill document inside a sandboxed iframe, guaranteeing clean single-page printing.

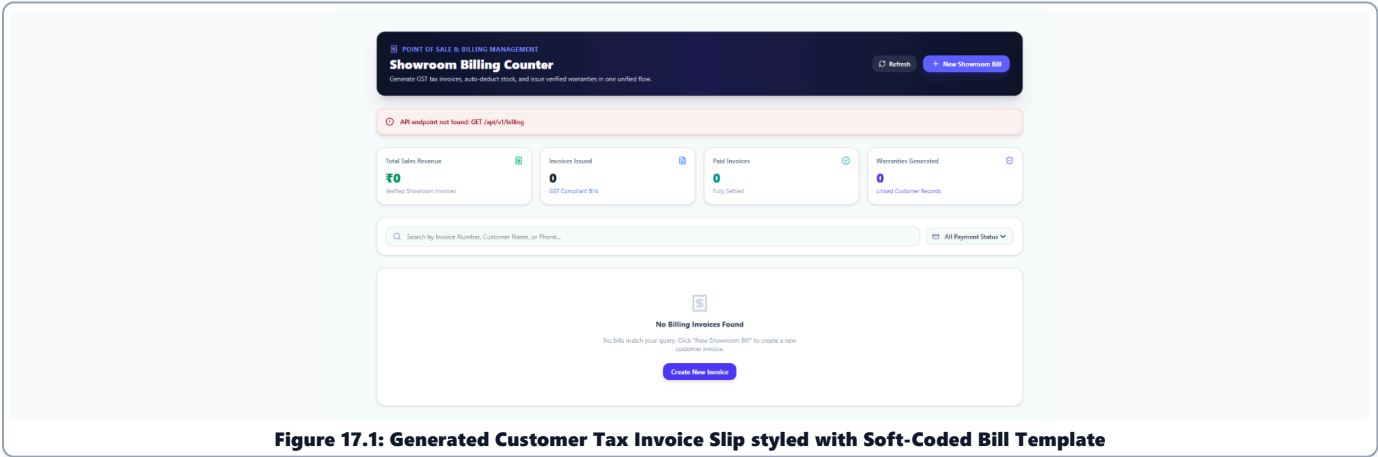


Figure 17.1: Generated Customer Tax Invoice Slip styled with Soft-Coded Bill Template

Template Property	Configuration Schema	Rendering Output
Theme Primary Color	Hex Color (e.g. #059669 or #4f46e5)	Border accents, headers, badge highlights
Watermark Matrix	Opacity text (e.g. "OFFICIAL INVOICE")	Diagonal semi-transparent background watermark
Signatory Block	Signatory title, plant location	Official stamp & authorized signature line

PAGE 18 — WAREHOUSE STOCK & INVENTORY EXPLORER

18.1 Warehouse Inventory Architecture

The inventory engine (backend/src/models/Stock.ts) tracks catalog products, serialized battery packs, and movement ledgers across warehouses and showrooms. Stock items maintain status flags (In Stock, Sold, Reserved, Damaged) and record comprehensive movement histories.

Stock Classification & Tracking:

- Battery Packs:** Serialized LFP / Li-ion models (48V, 60V, 72V).
- EV Scooters:** Chassis serialized vehicle inventory.
- Spare Parts & Chargers:** Batch tracked inventory units.
- Location Nodes:** Kota Plant, Showroom Counter, Rental Hub.

Stock Movement Audit Trail:

Every inventory operation logs a record in StockMovement.ts recording movement type (*Received, Sold, Issued, Returned, Transferred, Adjustment*), quantity, staff name, and timestamp.

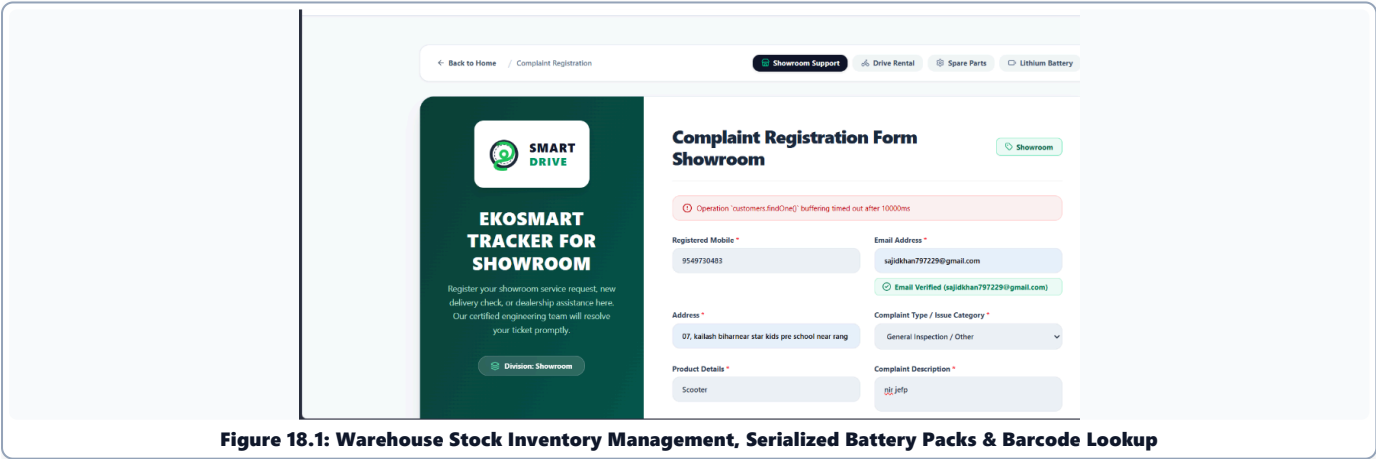


Figure 18.1: Warehouse Stock Inventory Management, Serialized Battery Packs & Barcode Lookup

Field Key	Data Type	Description / Operational Purpose
productId	String (Indexed)	Catalog SKU identifier (e.g. BAT-6030)
batterySerialNumber	String (Unique)	Laser-engraved physical battery pack serial number
quantity, unitPrice	Number	Available units in warehouse & unit selling price (₹)
status	Enum	In Stock, Sold, Reserved, Damaged

PAGE 19 — HTML5 CAMERA BARCODE & QR SCANNER MODULE

19.1 Real-Time Camera Barcode & QR Scanner Integration

Integrated via ScannerModal.tsx using @zxing/browser, the scanner leverages device webcams or mobile cameras to read Code 128, Code 39, EAN-13, and QR barcodes on physical battery packs, instantly auto-populating line items in POS billing.

Scanner Technical Capabilities:

- **HTML5 Stream:** Accesses rear mobile camera or USB webcam.
- **Multi-Format:** Decodes linear barcodes and 2D QR matrix.
- **Auto-Lookup:** Queries GET /stock/serial/:serial instantly.
- **Manual Fallback:** Allows direct keyboard typing if camera blocked.

POS Billing + Scanner Workflow:

During showroom billing, staff click "Scan Barcode". Pointing the camera at the battery laser serial instantly populates the product name, unit price, and serial number into the POS bill, eliminating typing mistakes.

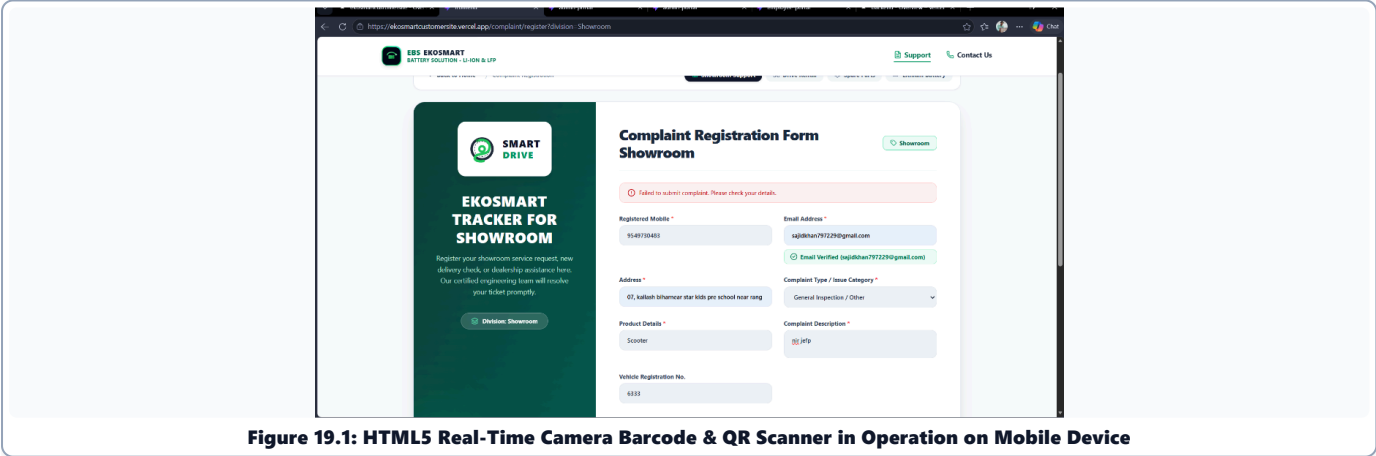


Figure 19.1: HTML5 Real-Time Camera Barcode & QR Scanner in Operation on Mobile Device

Step	Workflow Phase	System Action / Data Mutation
1	Barcode Scan	Staff scans battery serial; system verifies status === 'In Stock'.
2	POS Invoice Creation	Item added to bill; GST (18%) and grand total computed.
3	Database Sync	Stock.quantity decremented; StockMovement logged as 'Sold'.
4	Warranty Auto-Creation	Warranty record generated linking serial number to customer bill.

PAGE 20 — EMPLOYEE PORTAL: WORKSPACE & ASSIGNED QUEUE

20.1 Employee Workspace Architecture

The Employee Portal (employee-portal/) provides field engineers and showroom staff with a dedicated workstation. Authentication via POST /api/v1/auth/login verifies credentials, issues an employee_token, and evaluates RBAC permissions to dynamically configure the workspace.

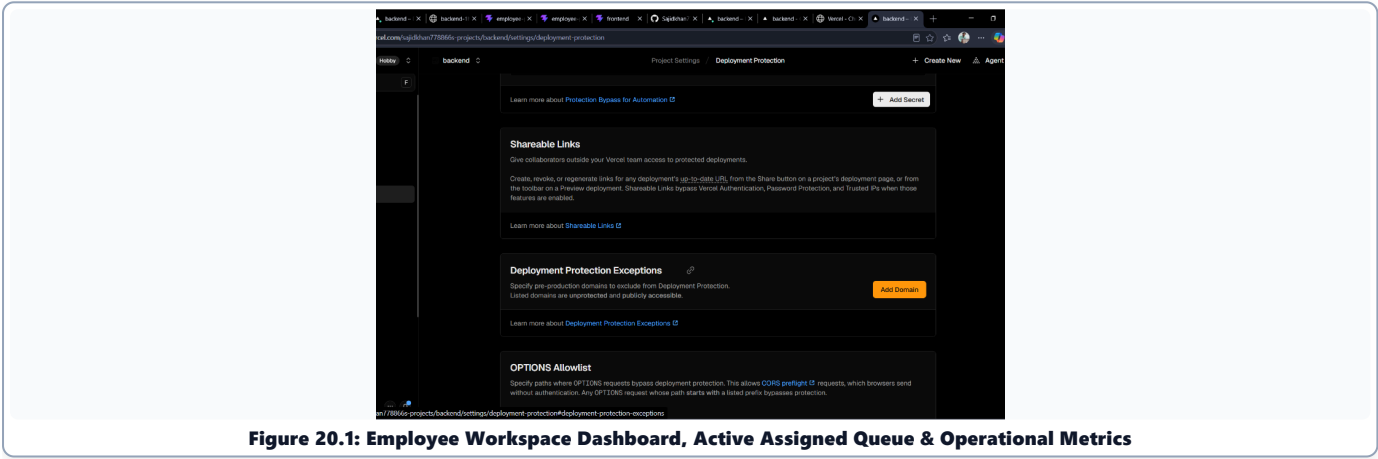


Figure 20.1: Employee Workspace Dashboard, Active Assigned Queue & Operational Metrics

20.2 Employee Portal Functional Module Directory

Module Page	Route	RBAC Permission	Operational Function
Workspace Dashboard	/	Authenticated Staff	Daily assigned queue summary, task status overview.
Assigned Complaints	/complaints	complaints	Inspect problem dossiers, update status, add repair notes.
Customer Directory	/customers	customers	Search customer profiles and linked service histories.
Warranty Terminal	/warranty	warranty	Check warranty coverage by serial, process service claims.
Showroom Billing	/billing	billing	Generate customer invoices, scan serials, compute GST.
Stock Inventory	/stock	stock	Check warehouse stock levels, scan barcodes, export CSV.
Staff Reports	/reports	reports	Export strictly assigned complaints to CSV.
My Salary Slip	/my-salary	payslips	View monthly remuneration & print official single-page slip.

PAGE 21 — EMPLOYEE COMPLAINT, CUSTOMER & WARRANTY WORKFLOW

21.1 End-to-End Service Resolution Workflow

When an employee logs into the portal, tickets assigned to their division appear in their active queue. Staff open the technical problem dossier (TicketDetailModal.tsx) to inspect customer complaints, photos, and dynamic test metrics.

[Assigned Ticket in Queue] --> [Inspect Customer Details & Photos] --> [Update Status to "IN PROGRESS"] | [Ticket Moved to Permanent History Ledger] <-- [Resolve Ticket & Add Remarks] <--+

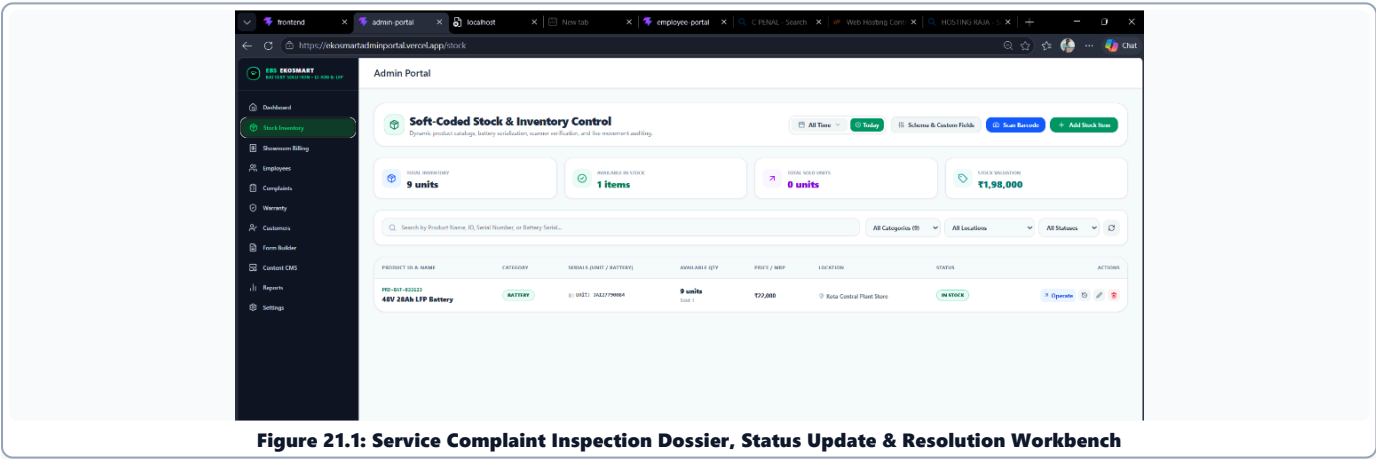


Figure 21.1: Service Complaint Inspection Dossier, Status Update & Resolution Workbench

21.2 Workload Isolation & Active Queue Rules

Operation / Rule	Technical Mechanism	Audit & Ledger Outcome
Assigned Queue Filter	Matches c.assignedTo with user.employeeId and user._id	Staff view only their assigned workload.
Diagnostic Progression	PATCH /api/v1/complaints/:id/status	Updates status from Pending to In Progress to Resolved.
Active vs. History	Resolved/Closed tickets clear from active list	Permanently archived in database for SLA reports.
Warranty Verification	Serial lookup against purchase date	Confirms active coverage before processing free repairs.

PAGE 22 — ENTERPRISE REPORTS, FILTERS & EXCEL EXPORT

22.1 Master Reporting & Analytics Engine

The reporting engine (admin-portal/src/pages/Reports.tsx & employee-portal/src/pages/Reports.tsx) provides analytics across complaints, warranties, inventory, and revenue. It supports division filters (*Battery, Rental, Showroom, Spare Parts*) and date presets.

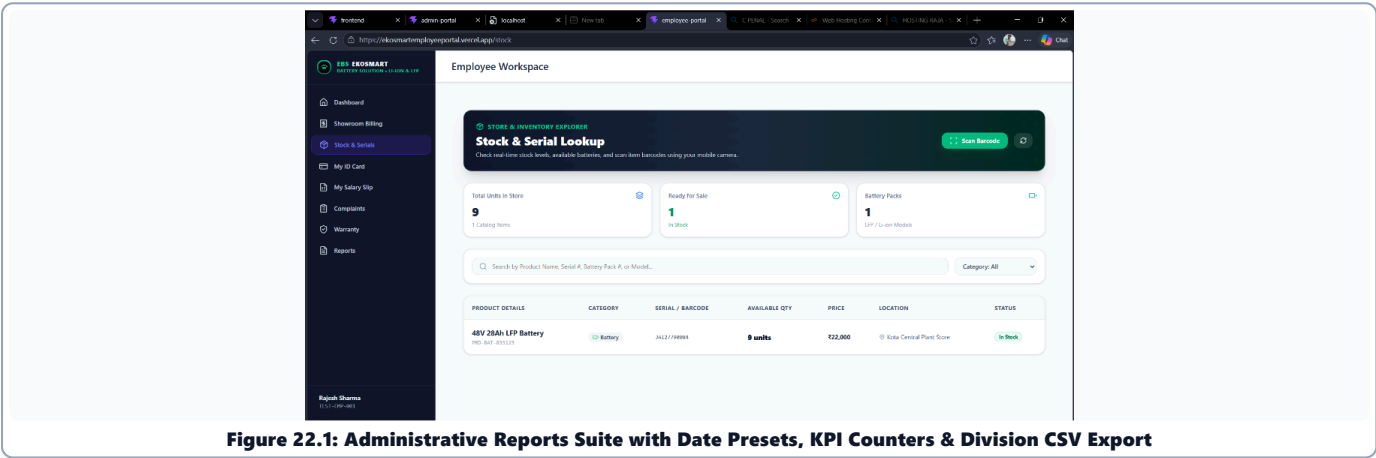


Figure 22.1: Administrative Reports Suite with Date Presets, KPI Counters & Division CSV Export

Date Range Filter Presets:

- Today (Default view) & Yesterday
- Last 7 Days & Last 30 Days
- This Month & All Time
- Custom Date Range (Start Date → End Date)

Workload Isolation Guarantee:

Employee Reports strictly filter tickets assigned to the logged-in staff member by matching `employeeId`, `_id`, `name`, and `email`, preventing data leakage across staff.

22.2 Excel / CSV Export Specifications

Export routines format data into RFC 4180 compliant CSV files with `\uFEFF` UTF-8 Byte Order Marks (BOM), ensuring multilingual customer names and Indian Rupee (₹8377:) currency symbols render cleanly in Microsoft Excel and Google Sheets.

PAGE 23 — GLOBAL DATE RANGE FILTERING & PRESETS ENGINE

23.1 Global Date Filtering Architecture

The date filtering module (`DateRangeFilter.tsx` & `backend/src/utlis/dateRange.ts`) provides unified temporal filtering across the platform. Present in Dashboard, Complaints, Warranty, Billing, Stock, and Reports, it guarantees synchronized date queries.

Unified Filter Presets:

- Today (Default):** 00:00:00 to 23:59:59 of current local date.
- Yesterday:** Past 24 hours snapshot.
- Last 7 Days / 30 Days:** Rolling weekly & monthly windows.
- This Month:** Calendar month start to current timestamp.
- Custom Range:** Exact user-specified date parameters.

Backend Date Resolution:

The backend helper `getDateRangeQuery()` translates preset query parameters into MongoDB query operators (`$gte` and `$lte` on `createdAt` or `purchaseDate`), ensuring index optimization.

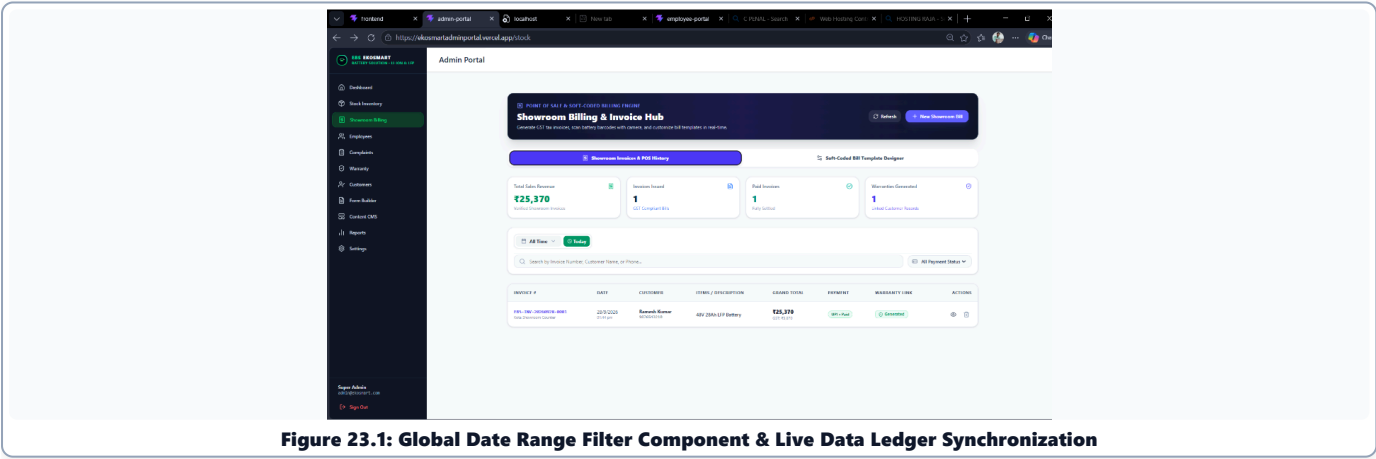


Figure 23.1: Global Date Range Filter Component & Live Data Ledger Synchronization

Module	Component Path	Supported Presets	Export Linkage
Dashboard	Dashboard.tsx	Today, 7D, 30D, Month, Custom	Updates all KPI counters
Warranty	Warranty.tsx	Today, 7D, 30D, Month, Custom	Filters warranties & CSV export
Showroom Billing	Billing.tsx	Today, 7D, 30D, Month, Custom	Filters invoices & sales CSV
Stock Inventory	Stock.tsx	Today, 7D, 30D, Month, Custom	Filters stock records & CSV

PAGE 24 — RESTFUL API DICTIONARY & DATABASE SCHEMA

24.1 RESTful API Endpoints Specification

The backend architecture mounts 12 route modules supporting both /api/v1/* and /api/* prefixes with JWT authentication, role guards, and serverless database auto-reconnection.

Module	Method & Endpoint	Auth / RBAC Guard	Operational Action
Auth	POST /api/v1/auth/login	Public	Validates credentials, returns JWT token
Complaints	POST /api/v1/complaints/public	Public	Creates service ticket (EBS-YYMM-XXXX)
Complaints	GET /api/v1/complaints/track/:ref	Public	Tracks status by ticket # or mobile
Complaints	PATCH /api/v1/complaints/:id/status	JWT (Staff)	Updates status & diagnostic remarks
Warranty	GET /api/v1/warranty/check/:query	Public	Checks warranty coverage by serial #
Warranty	POST /api/v1/warranty/register	Public	Registers warranty from showroom invoice
Billing	POST /api/v1/billing	JWT (Staff)	Generates POS bill & decrements stock
Stock	GET /api/v1/stock	JWT (Staff)	Lists warehouse stock with date filters
Employees	POST /api/v1/employees/:id/payslips	JWT (Admin)	Generates monthly payslip snapshot
Content (CMS)	PUT /api/v1/content/admin/update	JWT (Admin)	Updates dynamic CMS website copy

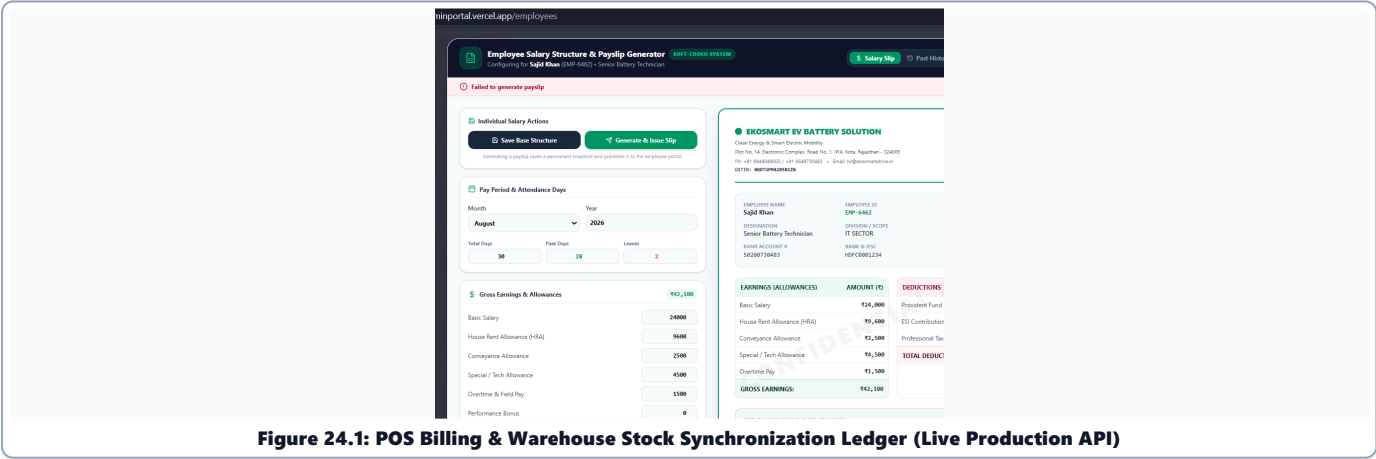


Figure 24.1: POS Billing & Warehouse Stock Synchronization Ledger (Live Production API)

24.2 MongoDB Collections Architecture

Collections: users, employees, complaints, complaintforms, warranties, customers, bills, billtemplates, stocks, stockmovements, payrollrecords, contents.

PAGE 25 — MASTER END-TO-END FLOW & IMPLEMENTATION MATRIX

25.1 Master End-to-End Operational Flow

[CUSTOMER] --> Public Portal --> Dynamic Form Intake --> Express API --> MongoDB Atlas (Ticket Created) [ADMIN] --> Admin Portal --> CMS / Form Designer / RBAC / Payroll --> MongoDB Atlas (Global Config) [EMPLOYEE] --> Employee Portal --> Diagnostic Dossier / POS / Stock --> MongoDB Atlas (Task Completed)

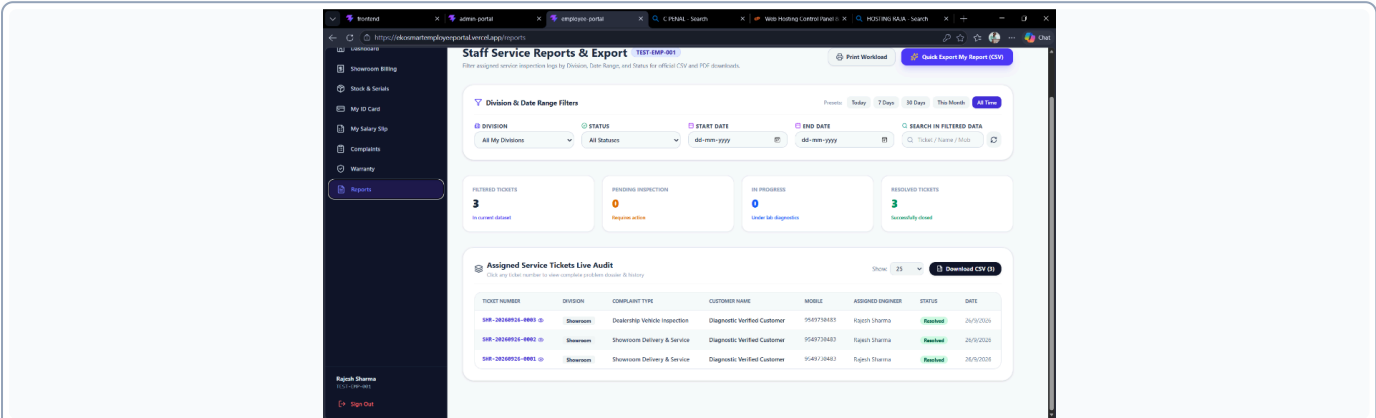


Figure 25.1: Staff Assigned Workload Live Audit, Date Filter Presets & Excel CSV Export Terminal

25.2 Comprehensive System Implementation Audit Matrix

System Module	Verified Status	Technical Architecture & Verification Notes
Public Website	IMPLEMENTED	React 19 SPA, dynamic CMS hero, soft-coded division cards, mobile menu
Admin Portal	IMPLEMENTED	KPI dashboard, CMS studio, dynamic form builder, staff management
Employee Portal	IMPLEMENTED	Role-gated workspace, assigned queue, POS billing, stock lookup
Backend API	IMPLEMENTED	Node.js, Express, TypeScript, dual prefix routing (/api/v1 & /api)
Database Layer	IMPLEMENTED	MongoDB Atlas, Mongoose ODM schemas, serverless reconnect middleware
Authentication & RBAC	IMPLEMENTED	JWT Bearer token, bcrypt password hashing, granular permission flags
Soft-Coded CMS	IMPLEMENTED	Database-driven hero, branding, dynamic service cards, legal policies
Dynamic Form Builder	IMPLEMENTED	Configurable field schema per division with live public form rendering
Complaint System	IMPLEMENTED	Automatic ticket generation (EBS-YYMM-XXXX), lifecycle tracking
Warranty Engine	IMPLEMENTED	Serial registration, automated duration expiry calculation, live search
Customer Directory	IMPLEMENTED	Central customer directory auto-ingested from complaints & invoices
Staff & Permissions	IMPLEMENTED	Staff CRUD, designation/department tracking, granular RBAC flags
Digital ID Cards	IMPLEMENTED	Printable ID card badge with verifiable QR badge and company branding
Salary & Payroll	IMPLEMENTED	Salary structure configuration, allowances, deductions, words INR
Isolated Payslip Print	IMPLEMENTED	Hidden iframe print engine generating single-page A4 payslips / PDF
Showroom POS Billing	IMPLEMENTED	Multi-item invoicing, GST 18%, sequential invoice numbering (INV-XXXX)
Warehouse Stock	IMPLEMENTED	Serialized battery tracking, inventory movement ledger, CSV export
Barcode Scanner	IMPLEMENTED	HTML5 camera scanner (@zxing/browser) with manual entry fallback
Reports & CSV Export	IMPLEMENTED	Multi-tab analytical reports, UTF-8 BOM CSV export, workload isolation
Date Range Filters	IMPLEMENTED	Global preset engine (Today, 7 Days, 30 Days, Month, Custom)
OTP SMS Verification	FRONTEND/UI ONLY	OTP UI dialog present; third-party SMS gateway not connected